

MARC THATCHER

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Research Profile

Programming-language theorist working on formal semantics, logic, type systems and graph rewriting. PhD research developed a high-level functional language for interaction nets, including an MLL-inspired type system, formal operational and observational semantics, and a prototype compiler and type checker in Haskell.

Research Interests

Programming language theory; formal semantics; logic; type systems; graph rewriting.

Education

University of Sussex

2023–2026

PhD, Computer Science

Thesis: *Designing a High-Level Language for Interaction Nets*

- Developed a typed functional-style calculus and programming language for a Turing-complete subset of interaction nets, with formal operational and observational semantics and an MLL-inspired type system.
- Implemented a prototype compiler and type checker in Haskell.
- Defined *productivity* for such nets and proved decidability results for a large subclass.

Thesis examined February 2026; awarded subject to minor corrections. Final corrected thesis accepted April 2026.

The Wharton School, University of Pennsylvania

2003–2005

MBA

Palmer Scholar (top 5% overall grade); Joseph Wharton Fellowship (merit-based scholarship); Omnicom Fellow; GPA: 4.0.

University of Manchester

1991–1994

BSc Computer Science, First Class Honours

Kilburn Prize for highest overall grade.

Research Publications

A Programming Language for Interaction Nets.

Termgraph 2026; full version October 2026. [Extended abstract link](#)

Designing a High-Level Language for Interaction Nets.

PhD thesis, University of Sussex, 2026. [Thesis link](#)

Research Talks

July 2026 Function-Constructor Nets and their Semantics, Termgraph; ISCTE, Portugal.

July 2026 A Programming Language for Interaction Nets, 1st Workshop on DIAGrams in LOGic and COMpu-tation; ISCTE, Portugal.

June 2026 The Problem of Observable Value in Interaction Nets, Southern and Midlands Logic Seminar; UCL.

May 2026 Computing with Infinite Interaction Nets: Continuity, Computability, Constructivity Workshop (on-line); Kyoto University, Japan.

November 2025 Interaction Net Productivity; University of Sussex.

April 2024 Parallel Functional Programming with Interaction Nets, British Colloquium for Theoretical Computer Science; University of Bath.

November 2023 Parallel Functional Programming with Interaction Nets, S-REPL; University of Bristol.

October 2023 Functional Interaction Nets; University of Sussex.

July 2023 A Functional Programming Language for Interaction Nets, International Workshop on Graph Computation Models; University of Leicester.

[Slides link](#)

Doctoral Training

Midlands Graduate School in the Foundations of Computing Science

April 2023

Courses in category theory, type theory, call-by-push-value and domain theory.

Teaching Experience

University of Sussex – Computer Science Tutor

2026–Present

International Study Centre

- Data Structures and Algorithms using Java (first-year undergraduate class).
- Further Programming and Professional Skills for Computing (teaching cover).

University of Sussex – Doctoral Tutor

2023–2026

Department of Informatics

- Limits of Computation (final-year undergraduates).
- Comparative Programming (final-year undergraduates; Haskell and Prolog).
- Compilers and Computer Architecture (second-year undergraduates).

Associate Fellow, Advance HE

November 2024

Undergraduate Summer Research Project Mentor

Summer 2023

Co-supervised a project on reversible computation using Geometry of Interaction. Supervisee is now a PhD candidate at the University of Sussex.

Technical Skills

Programming languages: Haskell, Python, Java, C, Standard ML, Scheme, Prolog.

Professional Experience

Chief Financial Officer, Hurbanot Limited (Part-time)

2020–Present

Chief Financial Officer, Assos of Switzerland

2017–2020

Director, Investment Banking, Kleinwort Benson

2013–2016

Director, Fixed Income Trading, Macquarie Bank

2008–2013

Distressed Debt Analyst, Davidson Kempner Capital Management

2004–2008

Assistant Director, N M Rothschild & Sons Limited

1994–2003